# Digital Ecosystems Architecture and the Unified Architecture Framework

## *Why a Coherence Framework Sits Above a Description Framework, and Depends On It*

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*Working Paper, Application Layer. Companion to Digital Ecosystems Architecture: A Three-Baseline Framework for Coherent Digital Realization.*

**Epistemic status:** *Positioning argument. Standards claims are anchored to UAF 1.2 (formal, July 2022) and ISO/IEC 19540-1:2022 at HIGH confidence from primary OMG sources. Claims that would depend on the full ISO/IEC/IEEE 42010:2022 normative text are marked MEDIUM and flagged for verification against the purchased standard.*

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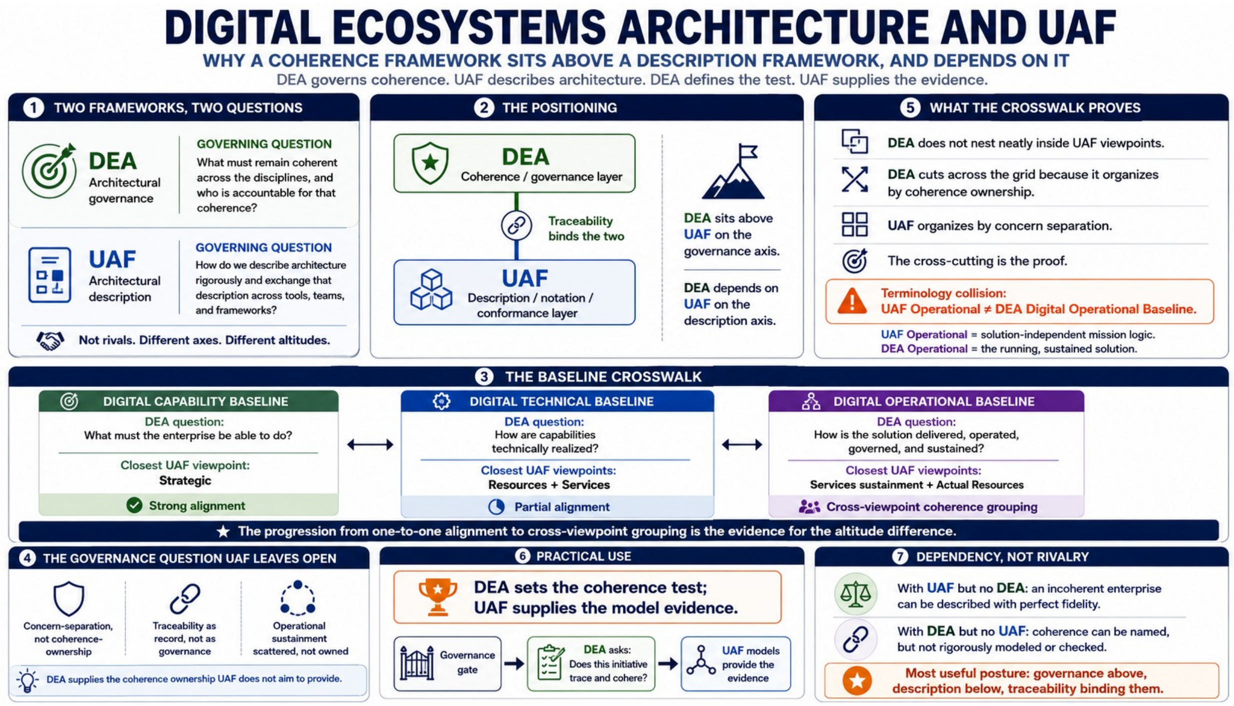
## Summary

The Unified Architecture Framework is a mature, internationally ratified standard for describing enterprise and system-of-systems architectures. It is rigorous, exchangeable between tools, and conformant to a recognized architecture-description standard. It is also, by its own account, a union of concerns inherited from its donor frameworks rather than a framework organized around architectural coherence or accountability. That is not a deficiency. It is a scope decision, and it leaves a specific question unanswered: across the disciplines that produce these descriptions, what must remain coherent, and who is accountable for that coherence?

Digital Ecosystems Architecture answers that question. This paper sets the two frameworks in their correct relation. The claim is not that DEA supersedes UAF. The claim is that the two operate at different altitudes on different axes: UAF on the axis of architectural *description*, DEA on the axis of architectural *governance*. DEA sits above UAF on the governance axis, because it specifies the coherence that descriptions must preserve and the accountability for preserving it. DEA depends on UAF on the description axis, because a coherence framework with no notation is an assertion. The relationship is dependency, not rivalry, and stating it precisely is what makes the positioning credible rather than grandiose. Put as plainly as the argument allows: DEA sets the coherence test; UAF supplies the model evidence.

## Figure 1. The Argument at a Glance




*A one-page summary of the positioning argument developed in this paper. The progression across the three baselines, from one-to-one alignment with a single UAF viewpoint to a coherence grouping that spans several, is the structural evidence for the altitude difference argued in Section 4. Confidence qualifications stated in the body, in particular the MEDIUM-confidence treatment of ISO/IEC/IEEE 42010:2022 in Section 5, are not reproduced in this figure; the figure should be read alongside the paper, not in place of it.*

# Contents

## 1. Two Frameworks, Two Questions

Enterprise and defense organizations that have invested in the Unified Architecture Framework sometimes encounter Digital Ecosystems Architecture and ask the natural question: is this a competitor, a replacement, or a layer on top? The question deserves a precise answer, because the wrong answer in either direction is costly. To treat DEA as a competitor to UAF is to set up a contest that misunderstands both. To treat DEA as merely a friendly summary of UAF is to discard the thing DEA actually contributes.

The frameworks answer different governing questions. UAF answers: *how do I describe this architecture rigorously, and exchange that description across tools, teams, and contributing frameworks?* Its concern is the fidelity and interoperability of architectural description. DEA answers: *what must remain coherent across the disciplines that produce these descriptions, and who is accountable for that coherence?* Its concern is the integrity of the whole as it crosses disciplinary boundaries.

These are not the same question with different vocabulary. A description can be flawless in UAF terms, every viewpoint populated, every model interchange-conformant, and the enterprise it describes can still be incoherent: capabilities that trace to no outcome, systems that satisfy no capability, solutions that are operationally unviable on arrival. UAF can faithfully model that incoherence. By itself, it does not make incoherence the governing problem, because its primary purpose is architectural description, not coherence accountability. That is the gap DEA is built to occupy.

## 2. What UAF Is, On Its Own Terms

A fair positioning argument begins by stating what the other framework does well, in its own language, without strawmanning. UAF is strong precisely where DEA is silent.

UAF is a domain metamodel organized as a two-dimensional grid. The rows are eleven viewpoints, each representing a set of stakeholder concerns: Architecture Management, Summary and Overview, Strategic, Operational, Services, Personnel, Resources, Security, Projects, Standards, and Actual Resources. The columns are aspects, the model kinds through which each viewpoint is examined: Taxonomy, Structure, Connectivity, Processes, States, Sequences, Information, Constraints, Roadmap, and Traceability. The intersection of a viewpoint and an aspect is a view specification, and the grid is the organizing artifact of the whole framework.

UAF carries genuine authority that DEA, as a conceptual base layer, does not. Its Domain Metamodel and Modeling Language are published as international standards, ISO/IEC 19540-1:2022 and ISO/IEC 19540-2:2022. It defines four graded types of conformance, from view-specification conformance through model-interchange conformance, so that a tool can make a precise and testable claim about how fully it implements the framework. It evolved from UPDM and consolidates the concerns of DoDAF, MODAF, and NAF, which means an architecture expressed in UAF can be exchanged with organizations and tools across the defense enterprise. And it is built and maintained by a consortium of the relevant institutions, with Northrop Grumman among the listed contributing companies to the current revision.

This is a serious instrument. Any framework claiming a position relative to UAF has to earn it against these strengths, not around them. DEA does not match UAF on any of these axes and does not attempt to. It has no metamodel of comparable formality, no ISO ratification, no conformance grades, no SysML implementation. If description, exchange, and conformance were the whole of the architecture problem, DEA would have nothing to add.

# 3. The Governance Question UAF Leaves Open

Description, exchange, and conformance are not the whole of the architecture problem, and UAF's own organizing principle makes that clear. The framework is candid about that principle, and that candor is the opening DEA occupies, an opening UAF leaves rather than overlooks.

The UAF grid is assembled from the concerns present in its donor frameworks. Its authors state plainly that the intent of the grid is not to be complete but to capture the information present in the contributing frameworks, and that as a consequence some gaps are evident. This is the decisive admission. UAF is a *union of inherited concerns*, refactored into a manageable and consistent grid. It is not organized around a thesis about what makes an enterprise architecture coherent, because coherence was not the design goal. Interoperable, rigorous, exchangeable description was.

Three consequences follow, and each marks ground that UAF leaves open.

## 3.1 Concern-separation, not coherence-ownership

UAF organizes by separation of concerns. The eleven viewpoints exist so that a security stakeholder, a personnel planner, and a systems engineer can each find the view that addresses their concern. This is the right organizing principle for description, and the wrong one for governance. Separation of concerns deliberately partitions the architecture so each stakeholder can reason locally. Coherence is the property that survives when those local views are recombined, and a framework organized around partitioning does not, by construction, foreground the recombination. DEA inverts the organizing principle: its three baselines are not concerns to be separated but coherence boundaries to be owned, each accountable for remaining consistent with the baselines adjacent to it.

## 3.2 Traceability as record, not as governance

UAF treats traceability as an aspect: a column in the grid, present in every viewpoint, defined as the mapping between elements in the architecture, between viewpoints and between domains. This is traceability as a kind of view, a thing you produce and consult. It is exactly the conception DEA argues against. In DEA, traceability is not a view among views; it is the mechanism by which governance operates at all. An initiative that cannot be traced from outcome to capability to system to solution to operation and back is not merely under-documented in DEA terms. It is ungoverned, because the governance function has no basis on which to decide. UAF gives you the apparatus to record a trace. DEA tells you that the trace is the precondition for the decision, and that its absence is a governance failure rather than a documentation gap.

### 3.3 Operational sustainment scattered, not owned

The concepts DEA gathers into its Digital Operational Baseline exist in UAF, but they are distributed across the grid by concern rather than gathered by accountability. Service levels and service contracts live in the Services viewpoint; fielded capabilities, achieved service levels, and actual resource configurations live in the Actual Resources viewpoint; the operational behavior that DEA would call sustainment is split between these and the solution-bearing Resources viewpoint. UAF has every element. What it does not have, because it is not its job, is a single accountable owner answerable for whether the solution is delivered, operated, governed, and sustained as a coherent whole. DEA supplies that owner. The elements are UAF's; the accountability is DEA's contribution.

## 4. The Altitude Argument: How the Baselines Cut Across the Viewpoints

The clearest evidence that DEA and UAF organize on different principles is what happens when you map one onto the other. DEA's three baselines do not nest neatly inside UAF's eleven viewpoints, one baseline per viewpoint or per clean group. They cut across the viewpoints. This is not a mapping failure to be smoothed over. It is the proof of the altitude difference, and it should be presented as such.

The mapping below is grounded in the normative element lists of the UAF 1.2 Domain Metamodel. It is offered at HIGH confidence for the viewpoint definitions and element memberships, which are drawn directly from the specification, and at MEDIUM confidence for the precise allocation of borderline elements, which is an interpretive judgment about DEA's intent rather than a fact about UAF.

| DEA Baseline | Primary DEA Question | Closest UAF Viewpoint Mapping | Positioning Implication |
|---|---|---|---|
| **Digital Capability** | What must the enterprise be able to do? | Strategic | Strong alignment. UAF Strategic is a natural descriptive home for capability modeling. |
| **Digital Technical** | How are capabilities technically realized? | Resources, plus the specification face of Services | Partial alignment. Technical realization crosses resource structure and reusable service specification. |
| **Digital Operational** | How is the solution delivered, operated, governed, and sustained? | Services sustainment elements, plus Actual Resources | Cross-viewpoint coherence grouping. Not equivalent to UAF Operational. |

*Table 1. DEA baselines mapped to their closest UAF 1.2 viewpoints. The progression from one-to-one alignment to cross-viewpoint grouping is itself the evidence for the altitude difference argued below.*

### 4.1 Digital Capability Baseline maps to Strategic

The cleanest correspondence. UAF's Strategic viewpoint is defined as the capability management process: the capability taxonomy, composition, dependencies, and evolution. That is very nearly a definition of what the Digital Capability Baseline owns. The baseline asks what the enterprise must be able to do; the Strategic viewpoint gives the notation for capabilities, their generalization hierarchy, their dependencies, and their phasing over time. Here DEA and UAF align almost one-to-one, and the Digital Capability Baseline can be operationalized largely within the Strategic viewpoint.

## 4.2 Digital Technical Baseline maps mainly to Resources, plus the specification face of Services

The Digital Technical Baseline asks how capabilities are technically realized. UAF's Resources viewpoint is defined as capturing a solution architecture consisting of resources, including software, systems, artifacts, capability configurations, and natural resources, that implement the operational requirements. That is the technical-realization concern. The reusable-service dimension of the technical baseline, the contracts and interfaces a capability exposes for others to consume, draws additionally on the specification face of the Services viewpoint, where service interfaces and service specifications are defined. The technical baseline therefore spans more than one UAF viewpoint even before reaching the operational seam.

## 4.3 Digital Operational Baseline spans Services sustainment and Actual Resources

This is the mapping the base layer flagged as hardest, and the metamodel resolves it. The Digital Operational Baseline does not correspond to UAF's *Operational* viewpoint. This is a genuine terminology collision and must be stated to avoid confusion: UAF's Operational viewpoint is defined as the logical architecture of the enterprise, describing operational behavior and exchanges required to support capabilities, explicitly in an implementation- and solution-independent manner. UAF *Operational* means solution-independent mission logic. DEA *Operational* means the running, sustained solution. They are nearly opposite ends of the realization spectrum.

DEA's Digital Operational Baseline maps instead to two places in the grid. Its sustainment-and-governance concern draws on the Services viewpoint, which carries service contracts, service policy, the governed-by relationship, and required and provided service levels. Its delivered-and-running concern draws on the Actual Resources viewpoint, defined as the analysis and verification of actual resource configurations and the illustration of expected or achieved configurations, and carrying fielded capability, actual services, and provided service levels as elements. The Digital Operational Baseline is therefore a coherence-grouping that deliberately spans the sustainment dimension of Services and the whole of Actual Resources. No single UAF viewpoint owns it, because UAF did not organize around the question the baseline answers.

## 4.4 What the crosswalk proves

Read down the mapping and the pattern is unmistakable. One DEA baseline aligns to one viewpoint; the next spans parts of two; the third spans the sustainment face of one viewpoint and the entirety of another, while explicitly not mapping to the viewpoint that shares its name. A framework that nested inside UAF would not behave this way. DEA cuts across the grid because it is organized by a different principle: coherence ownership rather than concern separation. The crosswalk is not the awkward part of the argument to be minimized. It is the argument.

# 5. The Shared Backbone, and an Honest Edition Gap

The two frameworks share a foundation, which is what makes their relationship a layering rather than a collision. Both ultimately answer to the same architecture-description standard, ISO/IEC/IEEE 42010, and that shared backbone is where DEA's traceability-as-governance claim could find its most rigorous home. One detail must be stated rather than hidden.

UAF conforms to the 2011 edition of the standard. The Domain Metamodel states that UAF conforms to the terms defined in ISO/IEC/IEEE 42010:2011 and that those terms form correspondence rules specified as constraints on UAF. DEA, as a contemporary framework, targets the 2022 edition. This is a real edition gap and is disclosed here deliberately. It matters because the 2022 edition develops the treatment of Correspondences and Correspondence Rules, the standard's mechanism for expressing and enforcing relations between architecture elements, including traceability relations. If, as appears to be the case, the 2022 edition gives Correspondence Rules the role of formally expressing and enforcing inter-element relations, then DEA's central claim, that traceability is governance rather than documentation, has a formal home in the conformance standard itself.

**Confidence note.** The 42010:2011 binding of UAF is confirmed at HIGH confidence from the primary UAF Domain Metamodel. The characterization of how 42010:*2022* treats Correspondences and Correspondence Rules is held at MEDIUM confidence, pending verification against the purchased normative text of the 2022 standard. This paragraph is the one place in the argument where a stronger claim is available but not yet grounded, and it is flagged here so the gap does not harden into settled fact. The positioning thesis does not depend on it; the thesis stands on the organizing-principle difference established in Sections 3 and 4. The 42010:2022 alignment, once verified, would strengthen Section 3.2 from a conceptual contrast to a standards-grounded one.

# 6. Dependency, Not Rivalry

The final move is the one that keeps the argument honest. Having claimed that DEA sits above UAF on the governance axis, the paper must state with equal clarity where DEA depends on UAF, because a coherence framework that disowned its need for notation would be making exactly the empty gesture it accuses lesser frameworks of.

DEA is a conceptual framework. It specifies what coherence to preserve, which boundaries to own, and what traceability must hold. It does not specify a notation in which to express any of this, and it deliberately does not, because committing to a notation at the conceptual layer would couple the framework to one tooling choice and degrade its relationships to whatever that notation could represent. This is not a temporary omission. It is a design commitment. The base layer was tested against a model-notation implementation and the implementation was rejected. Forcing the framework's relationships into a generic Association degraded them. The framework lives one level above instance-modeling notation. The same commitment runs in the other direction: DEA can be implemented through UAF views, and Section 4 shows where each baseline finds its descriptive home, but DEA should not be reduced to any one UAF profile, package, or viewpoint. A baseline that cuts across several viewpoints cannot be identified with any single one of them without losing the coherence grouping that is the point.

That design commitment is precisely why DEA needs UAF, or a framework like it. To operationalize a DEA baseline, an organization must eventually populate capabilities, model technical realization, and record actual configurations and service levels, and it must do so in a notation rigorous enough to be governed, exchanged, and conformance-checked. UAF is built for exactly this. The clean alignment of the Digital Capability Baseline to the Strategic viewpoint, and of the Digital Technical Baseline to the Resources viewpoint, is not a coincidence to be noted in passing. It is the seam along which DEA hands its governance intent down to UAF's descriptive machinery.

The relation, stated as plainly as it can be: DEA tells an organization what coherence to preserve and who owns it. UAF gives the organization the notation to model whether the coherence has in fact been preserved, and to exchange that model with everyone who needs it. Neither replaces the other. An organization with UAF and no governance framework can describe an incoherent enterprise in perfect fidelity. An organization with DEA and no description framework can name the coherence it wants and have no rigorous way to check whether it has it. The two are complementary by construction, and the layering runs in one direction: governance above, description below, traceability binding them.

## 7. Practical Use

For an organization that holds both frameworks, the division of labor reduces to a single operating rule: DEA sets the coherence test; UAF supplies the model evidence. DEA defines what must be true for an initiative to be coherent and governed, that it traces from outcome to capability to system to solution to operation and back, that each baseline is owned and accountable, that operational readiness was set as a constraint rather than discovered after delivery. UAF provides the rigorous, exchangeable, conformance-checkable models in which that trace is recorded and against which the coherence test is run.

In practice this means a governance gate asks the DEA question, does this initiative trace and cohere, and looks to the UAF models for the evidence to answer it. The capability lives in a Strategic view, the technical realization in Resources and Services views, the delivered and sustained solution in Services sustainment and Actual Resources views. DEA tells the gate what to check; UAF holds what gets checked. Neither function is optional, and neither substitutes for the other.

## 8. Conclusion

Digital Ecosystems Architecture earns a position above the Unified Architecture Framework, but only on a carefully specified axis and only at the cost of an equally explicit dependency. On the governance axis, DEA sits above UAF because it supplies what UAF, by its own design as a union of inherited concerns, does not: an organizing thesis about coherence and a structure of accountability for preserving it. On the description axis, DEA sits below UAF and depends on it, because conceptual governance without rigorous notation is only an intention.

The evidence for the altitude difference is structural, not rhetorical. DEA's three baselines cut across UAF's eleven viewpoints, aligning cleanly in one place, spanning two viewpoints in another, and explicitly declining to map to the viewpoint that shares the word *operational*. A framework that nested inside UAF would not behave this way. The cross-cutting is the proof. DEA organizes by coherence ownership; UAF organizes by concern separation; and the most useful thing an organization can do is run them together, with DEA naming the coherence and the accountability, and UAF carrying the description and the conformance.

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